ADADIVE

by THETA

A Decentralised Storage Economy on Cardano

WHITEPAPER

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This whitepaper is for informational purposes only and does not constitute financial, legal, investment, or tax advice. It is not an offer or solicitation to sell securities or any other regulated financial instrument in any jurisdiction.

THT (Theta Token) is a utility token designed to power transactions within the ADADIVE protocol. It is not designed to be, and should not be treated as, an investment vehicle. The information in this document is subject to change as the protocol develops, and ADADIVE makes no guarantees about future price, liquidity, or utility of THT.

ADADIVE is, at the time of writing, an early-stage protocol operating on the Cardano Preprod test network. Statements about future plans, including but not limited to mainnet launch dates, token economics, and protocol features, are forward-looking statements based on current intentions and are subject to change without notice.

Participation in any blockchain protocol, including ADADIVE, carries inherent risk, including but not limited to smart contract risk, regulatory risk, market risk, and the risk of total loss of funds. Readers should conduct their own research and consult independent professional advisors before making any decisions related to THT or the ADADIVE protocol.

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1. Executive Summary

ADADIVE is a decentralised storage economy built natively on the Cardano blockchain. The protocol enables a direct, peer-to-peer marketplace between storage providers — individuals or organisations with spare digital storage capacity — and users who need reliable, verifiable, censorship-resistant storage for their data.

The protocol is powered by THT (Theta Token), a native Cardano utility token that governs every economic interaction within ADADIVE: payment for storage, provider staking, deal collateral, and protocol governance.

Unlike centralised cloud storage providers, ADADIVE removes the single point of control, failure, and censorship inherent in services such as Amazon S3, Google Cloud Storage, and Microsoft Azure. Every storage deal on ADADIVE is recorded on the Cardano blockchain, with file content anchored via IPFS content identifiers (CIDs), creating permanent, publicly verifiable proof that data was stored as agreed.

As of June 2026, ADADIVE is live on the Cardano Preprod test network with real, measurable activity:

Metric	Value
Total storage capacity online	36 TB
Registered storage providers	5
Storage deals recorded on-chain	10 (6 active)
THT locked in active escrow	19,602.5 THT
THT exchange listing	Minswap DEX (ADA/THT pool)
External funding raised to date	$0 — fully bootstrapped
Time from inception to live testnet	Under 12 months

ADADIVE was conceived and built by two founders in Nigeria, with no external funding, demonstrating both the technical feasibility of the protocol and the capital efficiency of the team executing it.

2. Introduction

2.1 The Promise of Decentralisation

Since the launch of Bitcoin in 2009, blockchain technology has promised a new model for coordinating economic activity without centralised intermediaries. Decentralised finance (DeFi) has proven this thesis for money and lending. Decentralised storage extends the same thesis to data — one of the most fundamental resources of the digital economy.

Yet despite the maturity of blockchain infrastructure broadly, the Cardano ecosystem — one of the most actively developed and academically rigorous blockchain platforms in existence — has lacked a native, purpose-built decentralised storage protocol. ADADIVE was created to fill this gap.

2.2 Why Cardano

Cardano was selected as ADADIVE's foundation for several reasons:

- **Proof-of-Stake efficiency:** Cardano's Ouroboros consensus mechanism provides high security with significantly lower energy consumption than Proof-of-Work alternatives, aligning with sustainable infrastructure goals.
- **Native multi-asset support:** Cardano allows the creation of native tokens (such as THT) without the need for separate smart contract standards, reducing complexity and gas overhead.
- **Strong African ecosystem presence:** Cardano has made substantial commitments to African blockchain development, including partnerships in identity, education, and agriculture, creating a receptive ecosystem for an African-built protocol.
- **Deterministic, predictable fees:** Unlike fee markets that fluctuate unpredictably under network congestion, Cardano's fee structure is more predictable, which is essential for a protocol built around recurring, automated micro-payments per storage epoch.

2.3 Document Purpose

This whitepaper describes the problem ADADIVE addresses, the technical architecture of the protocol, the economics of the THT token, and the roadmap for development from the current testnet phase through to mainnet launch and long-term protocol governance.

3. The Problem

3.1 Centralisation of Digital Storage

The global data storage market is dominated by a small number of large cloud providers. This concentration creates several structural problems:

1. **Single points of failure:** Centralised storage providers represent a single point of failure. Outages, even at major providers, have repeatedly caused widespread service disruption across the internet.
2. **Censorship and deplatforming risk:** Centralised providers can and do remove content, suspend accounts, or restrict access based on internal policy, government pressure, or commercial considerations — without the data owner's consent.
3. **Price control:** With limited competition at the infrastructure layer, large providers set pricing with little competitive pressure, particularly affecting users and businesses in emerging markets who are price-sensitive.
4. **Lack of verifiability:** Users have no cryptographic proof that their data is being stored as agreed. They must trust the provider's word and service-level agreement, with limited recourse if that trust is broken.

3.2 The Cardano Storage Gap

As of early 2026, over 1,200 projects have been built on or are building on the Cardano blockchain, spanning DeFi, NFTs, identity, and real-world asset tokenisation. Each of these projects, at some point, needs to store data — metadata, media files, legal documents, user-generated content — outside the blockchain itself, since storing large files directly on-chain is prohibitively expensive.

Despite this clear and growing need, no protocol native to Cardano has offered a decentralised storage solution comparable to Filecoin or Arweave on other chains. Cardano projects needing decentralised storage have had to either rely on centralised providers (undermining the decentralisation thesis of their own products) or integrate separately with storage networks on entirely different blockchains, introducing additional complexity, bridging risk, and fragmentation.

3.3 The Emerging Market Access Problem

Existing decentralised storage protocols, while technically sound, were largely designed without emerging market constraints in mind. Minimum deal sizes, complex tooling, and pricing denominated in volatile or high-value tokens create friction for builders and users in regions such as Africa, where ADADIVE's founding team is based.

> *ADADIVE was built specifically to be accessible: low-friction onboarding, flexible deal sizes, and a token economy designed around the realities of emerging market participation.*

4. The ADADIVE Solution

ADADIVE is a peer-to-peer storage marketplace, built on Cardano, that connects storage providers directly with users, with all economic activity denominated in and settled through THT.

4.1 Core Principles

- **Trustless verification:** Every storage commitment is anchored on-chain. Proof of storage is verifiable by anyone, without needing to trust the provider's claims.
- **Economic alignment:** Providers are financially incentivised to store data reliably through staking and reward mechanisms, and financially penalised for failure through slashing.
- **Open participation:** Anyone with spare storage capacity and a Cardano wallet can become a provider. There are no centralised gatekeepers controlling who may participate.
- **Native Cardano integration:** ADADIVE does not bridge from another chain. It is built directly on Cardano, using Cardano-native tooling, wallets, and the THT native token.

4.2 How It Works — Overview

At a high level, ADADIVE coordinates four participants: storage providers, users (data owners), the Cardano blockchain (for settlement and verification), and the IPFS network (for the actual file storage layer).

5. **Provider registration:** A storage provider registers on ADADIVE, declaring available capacity and staking THT as a commitment bond.
6. **Deal creation:** A user selects a provider, specifies file(s), deal duration (in epochs), and locks the corresponding THT payment into a smart contract escrow.
7. **File pinning:** The file is uploaded and pinned to IPFS. The resulting content identifier (CID) — a unique cryptographic fingerprint of the file — is generated.
8. **On-chain anchoring:** The CID, along with deal metadata, is recorded in a Cardano transaction's metadata fields, creating a permanent, timestamped, publicly auditable record of the storage commitment.
9. **Epoch-based verification:** At regular intervals (epochs), the provider must demonstrate continued possession of the file. Successful proofs release a portion of the escrowed THT to the provider as payment.
10. **Settlement or slashing:** Upon successful completion of the deal duration, all escrowed THT is released to the provider. If the provider fails to provide proof of storage at any epoch, a portion of their staked THT is slashed as a penalty, and the affected portion of the deal is flagged for the user.

This document elaborates on each of these stages in the Protocol Architecture and Storage Deal Lifecycle sections below.

5. Protocol Architecture

5.1 System Components

ADADIVE consists of five primary technical components working in concert:

Component	Function
Cardano Smart Contracts	Manage THT escrow, staking, slashing, and deal state transitions on-chain
IPFS Pinning Layer	Stores and serves the actual file content; content-addressed via CIDs
Koios Query Layer	Reads Cardano blockchain state — wallet balances, transaction confirmations, metadata — without requiring a self-hosted node
Provider Network	The decentralised set of nodes offering storage capacity and responding to proof-of-storage challenges
Application Layer	The web interface (adadive.xyz) through which users and providers interact with the protocol

5.2 On-Chain Data Anchoring

ADADIVE makes deliberate use of Cardano's native transaction metadata fields rather than requiring custom on-chain storage of file content. Two metadata labels are used:

- **Label 1337 (ADADIVE protocol data):** Contains the structured deal record — CID, file metadata, provider and user addresses, deal duration, and pricing terms.
- **Label 721 (standard NFT-style metadata):** A widely recognised metadata standard in the Cardano ecosystem, included to ensure ADADIVE storage records are discoverable by existing Cardano metadata indexers and explorers.

This approach keeps on-chain footprint minimal (critical for transaction cost efficiency) while ensuring permanent, queryable proof of every storage commitment.

5.3 Wallet Integration

ADADIVE integrates with Cardano-compatible wallets, with Eternl as the primary supported wallet at the time of writing. Transaction construction is handled client-side using cardano-serialization-lib, ensuring users retain full custody of their private keys at all times. ADADIVE's application layer never has access to user private keys or seed phrases.

5.4 Storage Layer Flexibility

While IPFS (via Pinata as the initial pinning service) is the primary storage layer at launch, ADADIVE's architecture is designed to be storage-layer agnostic. Future protocol versions may support Arweave for permanent, pay-once storage use cases, and direct Cardano-native storage solutions as they mature, giving users a choice of storage layer per deal based on their durability and cost requirements.

6. THT Token Economics

6.1 Token Overview

Property	Value
Name	THT (Theta Token)
Blockchain	Cardano
Token Standard	Cardano Native Asset
Total Supply	100,000,000 THT (fixed)
Current Network	Preprod (testnet); Mainnet planned Q4 2026
DEX Listing	Minswap (ADA/THT pool)

6.2 Utility

THT is a multi-purpose utility token with the following functions within the protocol:

- **Storage payment:** All storage deals are denominated and paid for in THT. Users lock THT into escrow when creating a deal.
- **Provider staking:** Providers must stake THT to register on the network. Stake size influences eligible deal size and signals commitment.
- **Deal collateral:** A portion of each deal's value is held as collateral, which is forfeited (slashed) by the provider in the event of verified storage failure.
- **Governance:** THT holders will be able to vote on protocol parameters — fee structure, slashing severity, supported storage layers — as governance functionality is introduced (see Section 14).
- **Fee burning:** A portion of protocol fees collected from each completed deal is permanently removed from circulating supply, introducing deflationary pressure as network usage grows.

6.3 Supply Distribution

Allocation	Percentage	Amount	Purpose
Community Rewards	40%	40,000,000 THT	Provider epoch rewards, user incentives, ecosystem growth
Protocol Reserve	25%	25,000,000 THT	Long-term protocol development and operational runway

Team & Advisors	15%	15,000,000 THT	Founder and future team allocation, vested over 24 months
Ecosystem Grants	12%	12,000,000 THT	Third-party developer and integration grants
Initial Liquidity	8%	8,000,000 THT	DEX liquidity bootstrapping (Minswap ADA/THT pool)

6.4 Pricing Mechanism

Storage pricing on ADADIVE is denominated in THT per gigabyte per epoch, set independently by each provider within protocol-defined bounds. This rate is distinct from — and not to be confused with — the market exchange rate of THT against ADA or fiat currency, which fluctuates according to open market supply and demand on Minswap.

At the time of writing, the indicative average storage rate across active providers is 0.0013 THT per GB per epoch. This figure reflects provider-set storage pricing, not a token market valuation.

6.5 Current Liquidity Status

> *In the interest of transparency: THT's Minswap liquidity pool is in an early bootstrapping phase with minimal depth at the time of writing. Prospective participants should treat any current market price as indicative only and recognise the limitations of trading in a low-liquidity environment. ADADIVE's roadmap (Section 12) includes a dedicated phase for deepening liquidity ahead of and following mainnet launch.*

7. Provider Network & Reputation

7.1 Becoming a Provider

Storage providers are the backbone of the ADADIVE network. Any individual or organisation with available digital storage capacity and a Cardano wallet may apply to become a provider. The onboarding process, ahead of mainnet, consists of:

11. **Application:** Prospective providers submit their available capacity, region, and wallet address through the ADADIVE application portal.
12. **Review:** Applications are reviewed to confirm legitimacy and technical readiness ahead of testnet/mainnet onboarding.
13. **Staking:** Approved providers stake THT proportional to their intended storage commitment.
14. **Activation:** Once staked, the provider's capacity becomes available in the marketplace for users to create deals against.

7.2 Reputation Scoring

Each provider accumulates an on-chain reputation score based on their historical performance, calculated from:

- Successful epoch proofs delivered, as a proportion of total epochs served
- Total deal volume successfully completed
- Uptime and responsiveness to proof-of-storage challenges
- Time active on the network (longevity signal)

This reputation score is publicly visible on the ADADIVE network explorer, allowing users to make informed decisions about which providers to transact with, similar in spirit to seller ratings on traditional e-commerce marketplaces, but derived from verifiable on-chain data rather than self-reported reviews.

7.3 OG Provider Status

Providers who register and remain active during the testnet phase, ahead of mainnet launch, receive a permanent 'OG Provider' designation. This status is recorded on-chain and provides priority deal-matching visibility once the network scales, recognising early contributors to the network's bootstrapping phase.

8. Storage Deal Lifecycle

This section details the complete lifecycle of a storage deal, from initiation through to completion or failure.

8.1 Deal States

State	Description
Pending	Deal created, THT locked in escrow, awaiting on-chain confirmation
Active	Confirmed on-chain; storage is in progress; epoch proofs being submitted
Completed	All agreed epochs successfully fulfilled; full payment released to provider
Failed	Provider failed to deliver required proof of storage; partial slashing applied
Cancelled	Deal cancelled prior to on-chain confirmation; THT returned to user

8.2 Epoch-Based Proof of Storage

ADADIVE divides each storage deal's duration into discrete epochs. At each epoch boundary, the provider must submit proof that the file remains stored and retrievable. This is implemented by verifying the IPFS pin status of the relevant CID and cross-referencing it against the on-chain deal record.

Successful proof at an epoch triggers release of that epoch's proportional THT payment from escrow to the provider. This creates a continuous, automated payment stream tied directly to verified performance, rather than a single lump payment that could mask intermittent failures.

8.3 Failure Handling and Slashing

If a provider fails to deliver proof of storage at a given epoch, the following occurs:

15. **The failure is recorded against the provider's reputation score.**
16. **A pre-defined percentage of the provider's staked THT for that deal is slashed.**
17. **The user is notified and may choose to terminate the remaining deal and reclaim unused escrowed THT, or allow the provider an opportunity to restore service.**

Slashed THT is partially redirected to the protocol treasury and partially burned, depending on parameters set by protocol governance.

9. Security & Risk Management

9.1 Smart Contract Security

ADADIVE's smart contracts manage THT escrow, staking, and slashing logic — the components of the protocol carrying the highest financial risk if flawed. Prior to mainnet deployment, ADADIVE is committed to:

- Independent third-party smart contract security audit by a recognised Cardano-focused auditing firm
- Public disclosure of audit findings and remediation
- A phased mainnet rollout with deal size limits initially capped, increasing gradually as on-chain track record accumulates

9.2 Key Custody

ADADIVE never takes custody of user or provider private keys. All transaction signing occurs client-side through the user's own wallet (Eternl at present). This significantly reduces the attack surface compared to protocols that require custodial key management.

9.3 Infrastructure Security

ADADIVE's backend infrastructure, including credential management, input validation on all API endpoints, and database access controls, follows standard security best practices, including:

- Strict schema validation (via Zod) on every API input, rejecting malformed Cardano addresses, transaction hashes, and content identifiers before they reach any downstream system
- Environment-based secret management, with no credentials committed to version control
- Rate limiting on public-facing endpoints to mitigate abuse and denial-of-service risk

9.4 Known Risks

ADADIVE acknowledges the following categories of risk inherent to the protocol's current stage of development:

18. **Early-stage / testnet risk:** The protocol is currently operating on Cardano Preprod, not mainnet. Mainnet deployment timing depends on completion of security auditing and is not guaranteed to occur by any specific date.
19. **Liquidity risk:** THT currently has limited DEX liquidity, which can result in significant price volatility on small trade volumes.

20. **Provider concentration risk:** At the current network size (5 providers), the network has not yet achieved the geographic and operational diversification that characterises a mature decentralised storage network.
21. **Regulatory risk:** The regulatory treatment of utility tokens varies by jurisdiction and may change. ADADIVE will adapt its operations as needed to comply with applicable law in jurisdictions where it operates.

10. Market Opportunity

10.1 Global Decentralised Storage Market

The global decentralised storage market is projected to reach approximately $1.5 billion by 2028, driven by increasing demand for censorship-resistant, verifiable, and resilient data storage across Web3 applications, enterprise backup use cases, and digital archiving.

10.2 The Cardano-Specific Opportunity

With over 1,200 projects building on Cardano and no existing native decentralised storage protocol, ADADIVE occupies a structurally advantaged position: it is not competing for market share within an already-served niche, but rather establishing the storage layer for an entire blockchain ecosystem that currently lacks one.

10.3 DePIN — Decentralised Physical Infrastructure Networks

ADADIVE belongs to the broader category of Decentralised Physical Infrastructure Networks (DePIN) — protocols that coordinate real-world physical resources (in this case, storage hardware) through blockchain-based incentive mechanisms. DePIN is among the fastest-growing categories within Web3, and ADADIVE represents one of the first DePIN protocols built natively on Cardano.

10.4 Emerging Market Demand

African businesses and developers face disproportionately high costs and limited local infrastructure options for cloud storage. A protocol priced in a low-cost utility token, with flexible deal sizing and low onboarding friction, directly addresses a documented gap in infrastructure access for emerging market participants — both as storage consumers and as potential storage providers monetising otherwise idle capacity.

11. Competitive Landscape

ADADIVE operates within the decentralised storage category alongside several established protocols on other blockchains. The table below outlines key points of differentiation.

Protocol	Blockchain	Key Distinction from ADADIVE
Filecoin	Own L1 (Filecoin blockchain)	Mature, large-scale network; significantly higher technical complexity and minimum deal requirements
Arweave	Own L1 (Arweave blockchain)	Permanent pay-once model; no Cardano-native integration
Storj	Ethereum (for payments)	Hybrid centralised/decentralised model; not natively integrated with Cardano tooling or wallets
ADADIVE	Cardano	First decentralised storage protocol native to Cardano; flexible epoch-based deals; designed for emerging market accessibility

ADADIVE's core differentiation is not attempting to out-compete established multi-billion-dollar networks on raw scale, but to serve a market — the Cardano ecosystem and its emerging-market-heavy developer base — that existing protocols do not natively address.

12. Roadmap

Phase 1 — Foundation (Complete)

- Protocol architecture design and THT token creation on Cardano
- Storage deal smart contracts deployed to Preprod testnet
- Provider staking and registration system operational
- Eternl wallet integration
- Network explorer with live on-chain telemetry
- 5 providers onboarded; 36TB capacity online; 10 deals recorded; 19,602.5 THT in escrow
- THT listed on Minswap DEX

Phase 2 — Growth (In Progress, Q3 2026)

- Public website and THT token information page live at adadive.xyz
- Provider application portal for pre-mainnet onboarding
- Community building across Discord and social channels
- Publication of this whitepaper
- Pursuit of accelerator and grant support (Adaverse, Qatar Fintech Hub)
- Expansion to 10+ registered providers on testnet

Phase 3 — Mainnet Launch (Target Q4 2026)

- Independent smart contract security audit
- Cardano mainnet deployment
- THT liquidity deepening on Minswap
- Structured token distribution event for early community members
- Provider incentive programme activation
- Target: first 100 mainnet storage deals

Phase 4 — Scale (2027 and beyond)

- On-chain governance enabling THT holder voting on protocol parameters
- Public developer SDK for third-party integration
- Additional storage layer support (Arweave, others)
- Mobile application
- Enterprise-grade storage tier with service-level guarantees
- Transition toward full decentralised governance (DAO)

13. Team

Chima Ifeanyi Frank — CEO & Co-Founder

Full-stack developer and business strategist based in Enugu, Nigeria. Leads product strategy, tokenomics design, frontend engineering, and external partnerships for ADADIVE. Brings prior experience in Cardano token development from an earlier project, alongside an academic background in Medical Laboratory Science, contributing a methodical, evidence-based approach to protocol development and testing.

Chima Emmanuel — CTO & Co-Founder

Core protocol developer responsible for smart contract architecture, backend infrastructure, and Cardano blockchain integrations. Leads development of the storage deal contracts, provider staking mechanism, and the epoch-based proof verification system at the heart of the protocol.

> *ADADIVE has been built entirely by its two founders, with no external engineering hires and no external funding, demonstrating both technical capability and capital discipline.*

14. Governance

At launch, protocol parameters (fee rates, slashing severity, supported storage layers) are set by the founding team to ensure stability during the network's early growth phase. As the network matures and THT distribution broadens beyond the founding allocation, ADADIVE intends to transition toward on-chain governance, in which THT holders may propose and vote on:

- Protocol fee percentages
- Slashing parameters for provider non-performance
- Approval of new supported storage layers
- Allocation of the Protocol Reserve and Ecosystem Grants pools

The precise governance mechanism — including voting weight calculation, proposal thresholds, and execution timelocks — will be detailed in a dedicated governance specification ahead of its implementation, expected as part of Phase 4 of the roadmap.

15. Legal & Regulatory Considerations

ADADIVE is committed to operating in compliance with applicable laws in the jurisdictions in which it operates. The regulatory treatment of utility tokens, decentralised protocols, and blockchain-based services continues to evolve globally, and ADADIVE will adapt its structure and operations as needed in response to regulatory developments.

THT is designed and intended to function as a utility token providing access to and payment for services within the ADADIVE protocol. It is not designed to represent equity, profit-sharing rights, or any other form of security in any ADADIVE-affiliated entity. Prospective participants should independently assess the regulatory treatment of THT in their own jurisdiction before participating in the protocol.

ADADIVE is, at the time of writing, an early-stage, unincorporated technology project based in Nigeria. Formal incorporation is anticipated as part of the protocol's preparation for mainnet launch and institutional partnership development.

16. Conclusion

ADADIVE represents a focused, technically grounded answer to a clear gap in the Cardano ecosystem: the absence of a native, decentralised, verifiable storage layer. Built by two founders in Nigeria over the course of less than a year, with no external funding, the protocol has progressed from concept to a live, functioning testnet deployment with real providers, real storage deals, and real value secured in escrow.

The work ahead — security auditing, mainnet deployment, liquidity deepening, and progressive decentralisation of governance — is substantial, but the foundation has been built, tested, and proven to function as designed.

> *ADADIVE is not a proposal. It is a working protocol, built in public, with the numbers to show for it. The next chapter is mainnet.*